

Second Derivatives and Convex Function

Learning objectives

By the end of this chapter, you should be able to

- compute the Hessian matrix of a twice differentiable function;
- interpret Hessian quadratic forms as directional curvature and relate them to quadratic level curves;
- classify a symmetric matrix from its quadratic form, eigenvalues, or principal minors;
- use Taylor's theorem and the Hessian to establish convexity or concavity; and
- distinguish necessary, sufficient, and equivalent second-order conditions.

Let n be a positive integer, let $S \subseteq \mathbb{R}^n$ be open and convex, let $f : S \rightarrow \mathbb{R}$ be differentiable, and let $\mathbf{x}, \mathbf{y} \in S$. For such a function, convexity is equivalent to the tangent-plane inequality

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}).$$

Checking this inequality for every pair $\mathbf{x}, \mathbf{y}$ may be inconvenient. The Hessian turns the same question into a pointwise matrix test.

5.1 The Hessian matrix

Definition 5.1 (Hessian matrix). Let $S \subseteq \mathbb{R}^n$, let $f : S \rightarrow \mathbb{R}$, and let $\mathbf{x}$ be an interior point at which all second partial derivatives exist. The *Hessian* of f at $\mathbf{x}$ is

$$H_f(\mathbf{x}) := \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix}.$$

If the second partial derivatives are continuous near $\mathbf{x}$, Clairaut's theorem gives equality of the mixed partials, so $H_f(\mathbf{x})$ is symmetric.

Example 5.1. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) := x^3 + 2x + e^x$. Then

$$f'(x) = 3x^2 + 2 + e^x, \quad H_f(x) = f''(x) = 6x + e^x.$$

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x_1, x_2) := x_1^3 + 2x_1x_2 + x_2^2$. Then

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 3x_1^2 + 2x_2 \\ 2x_1 + 2x_2 \end{bmatrix}, \quad H_f(\mathbf{x}) = \begin{bmatrix} 6x_1 & 2 \\ 2 & 2 \end{bmatrix}.$$

5.2 Directional curvature and quadratic level curves

Let $S \subseteq \mathbb{R}^n$ be open, let $f : S \rightarrow \mathbb{R}$ have continuous second partial derivatives, let $\mathbf{x} \in S$, and define $\mathbf{d} := (d_1, \dots, d_n)^\top \in \mathbb{R}^n \setminus \{\mathbf{0}\}$. Since S is open, there is an $\varepsilon > 0$ such that $\mathbf{x} + t\mathbf{d} \in S$ whenever $|t| < \varepsilon$. Define

$$\phi : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}, \quad \phi(t) := f(\mathbf{x} + t\mathbf{d}).$$

The chain rule gives

$$\phi'(0) = \nabla f(\mathbf{x})^\top \mathbf{d}, \quad \phi''(0) = \mathbf{d}^\top H_f(\mathbf{x}) \mathbf{d}.$$

Thus $\mathbf{d}^\top H_f(\mathbf{x}) \mathbf{d}$ is the second directional derivative of f at $\mathbf{x}$ along $\mathbf{d}$. Its sign records upward, downward, or zero second-order curvature in that direction. A zero value means only that the quadratic term vanishes in that direction; higher-order variation may remain.

On an open convex set, nonnegativity of $\mathbf{d}^\top H_f(\mathbf{x}) \mathbf{d}$ for every point $\mathbf{x}$ and every direction $\mathbf{d}$ characterizes convexity, while strict positivity for every nonzero direction is sufficient for strict convexity. The negative analogues give the concavity statements. The precise result is stated in [Theorem 5.5](#); both universal quantifiers are essential.

Define $q_1, q_2, q_3 : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $q_1(\mathbf{x}) := \frac{1}{2}(x_1^2 + 4x_2^2)$, $q_2(\mathbf{x}) := \frac{1}{2}x_1^2$, and $q_3(\mathbf{x}) := \frac{1}{2}(x_1^2 - x_2^2)$. For $i \in \{1, 2, 3\}$ and $c \in \mathbb{R}$, the level set of q_i at level c is $\{\mathbf{x} \in \mathbb{R}^2 : q_i(\mathbf{x}) = c\}$. The plots below use equally spaced values of c , even though the level curves need not be equally spaced geometrically.

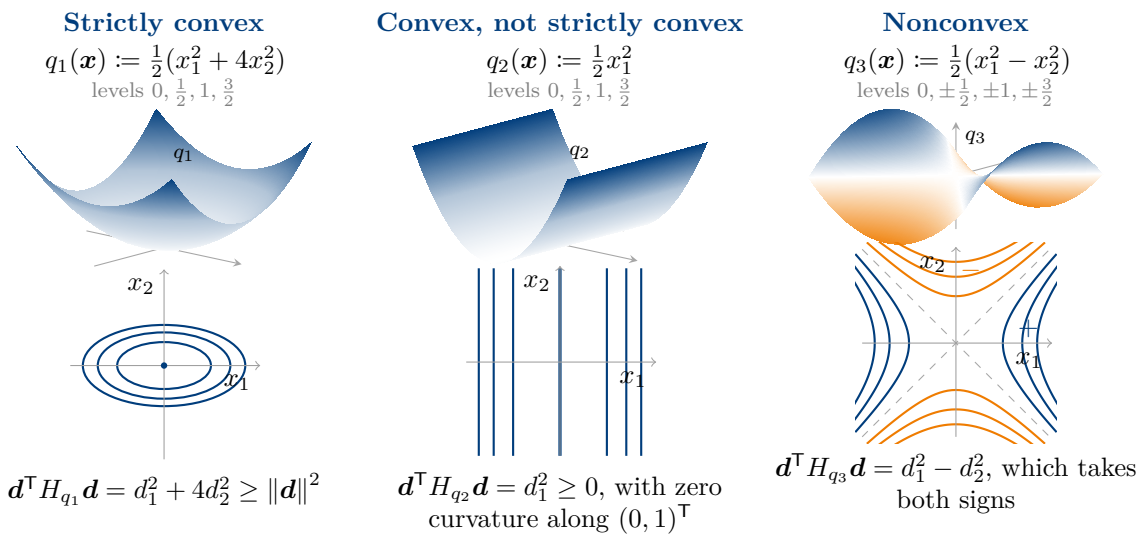


Figure 5.1: Three-dimensional graphs, directional curvature, and equally spaced level values of three quadratic functions.

5.3 Definiteness of symmetric matrices

The preceding directional-curvature formula motivates the following terminology.

Definition 5.2 (Definiteness). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric. It is

- *positive semidefinite*, written $\mathbf{A} \succeq 0$, if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for every $\mathbf{x} \in \mathbb{R}^n$;
- *positive definite*, written $\mathbf{A} \succ 0$, if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for every $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$;
- *negative semidefinite*, written $\mathbf{A} \preceq 0$, if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \leq 0$ for every $\mathbf{x} \in \mathbb{R}^n$;
- *negative definite*, written $\mathbf{A} \prec 0$, if $\mathbf{x}^\top \mathbf{A} \mathbf{x} < 0$ for every $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$; and
- *indefinite* if its quadratic form takes both positive and negative values.

Warning

The eigenvalue and Sylvester tests below require symmetry. For a nonsymmetric matrix $\mathbf{A}$, its quadratic form depends only on its symmetric part:

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \left(\frac{\mathbf{A} + \mathbf{A}^\top}{2} \right) \mathbf{x}.$$

Thus the eigenvalues or leading principal minors of $\mathbf{A}$ itself need not characterize the sign of its quadratic form.

Example 5.2. For $\mathbf{A}_1 := \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$,

$$\mathbf{x}^\top \mathbf{A}_1 \mathbf{x} = 2 \left(x_1 - \frac{x_2}{2} \right)^2 + \frac{5}{2} x_2^2 > 0 \quad (\mathbf{x} \neq \mathbf{0}),$$

so $\mathbf{A}_1 \succ 0$. For $\mathbf{A}_2 := \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$,

$$\mathbf{x}^\top \mathbf{A}_2 \mathbf{x} = (x_1 + 2x_2)^2 - 3x_2^2.$$

It equals 1 at $\mathbf{x}^+ := (1, 0)^\top$ and -3 at $\mathbf{x}^- := (-2, 1)^\top$, so $\mathbf{A}_2$ is indefinite.

5.3.1 The eigenvalue test

Theorem 5.1 (Eigenvalue test). Let $\mathbf{A}$ be real and symmetric. Then

$$\mathbf{A} \succeq 0 \iff \text{every eigenvalue is nonnegative,}$$

$$\mathbf{A} \succ 0 \iff \text{every eigenvalue is positive,}$$

$$\mathbf{A} \preceq 0 \iff \text{every eigenvalue is nonpositive,}$$

$$\mathbf{A} \prec 0 \iff \text{every eigenvalue is negative.}$$

Moreover, $\mathbf{A}$ is indefinite exactly when it has both a positive and a negative eigenvalue.

Proof. We prove the PSD statement. If $\mathbf{A} \succeq 0$, let λ be an eigenvalue with corresponding eigenvector $\mathbf{q} \neq \mathbf{0}$, so $\mathbf{A}\mathbf{q} = \lambda\mathbf{q}$. Then $0 \leq \mathbf{q}^\top \mathbf{A}\mathbf{q} = \lambda \|\mathbf{q}\|^2$, so $\lambda \geq 0$. Conversely, the spectral theorem supplies an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and eigenvalues $\lambda_1, \dots, \lambda_n$ such that $\mathbf{A} = \mathbf{Q} \operatorname{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}^\top$. With $\mathbf{y} := \mathbf{Q}^\top \mathbf{x}$,

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i=1}^n \lambda_i y_i^2 \geq 0.$$

Strict inequalities give the PD case; applying the result to $-\mathbf{A}$ gives the negative cases. $\square$

Example 5.3. The matrix $\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$ has eigenvalues 0, 5, so it is PSD but not PD. The matrix

$$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

has eigenvalues approximately 1, -2.3166 , 4.3166 , so it is indefinite.

5.3.2 Leading principal minors

Write $\mathbf{A} := [a_{ij}]_{i,j=1}^n$. For $k = 1, \dots, n$, set $\Delta_k := \det([a_{ij}]_{i,j=1}^k)$.

Theorem 5.2 (Sylvester's criterion). *For a real symmetric matrix $\mathbf{A}$,*

$$\mathbf{A} \succ 0 \iff \Delta_k > 0 \quad (k = 1, \dots, n),$$

and

$$\mathbf{A} \prec 0 \iff (-1)^k \Delta_k > 0 \quad (k = 1, \dots, n).$$

For a nonempty index set $I \subseteq \{1, \dots, n\}$, define the principal submatrix

$$\mathbf{A}[I] := [a_{ij}]_{i,j \in I}.$$

Its determinant $\det \mathbf{A}[I]$ is the associated *principal minor*. The leading principal minors correspond to $I = \{1, \dots, k\}$.

Theorem 5.3 (Principal-minor test for positive semidefiniteness). *Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric. Then*

$$\mathbf{A} \succeq 0 \iff \det \mathbf{A}[I] \geq 0 \quad \text{for every nonempty } I \subseteq \{1, \dots, n\}.$$

Warning

This form of Sylvester's criterion tests definiteness. Nonnegative leading principal minors alone do not imply positive semidefiniteness. For example, $\operatorname{diag}(0, -1)$ has both leading minors zero but is not PSD.

Example 5.4. The matrix $\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$ has leading minors 2, 3, 4, so it is PD. The

matrix $\begin{bmatrix} -1 & 1 & 1 \\ 1 & -4 & 0 \\ 1 & 0 & -2 \end{bmatrix}$ has leading minors $-1, 3, -2$, so it is ND.

5.4 Taylor's theorem in several variables

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, define the line segment joining them by

$$[\mathbf{x}, \mathbf{y}] := \{\mathbf{x} + t(\mathbf{y} - \mathbf{x}) : t \in [0, 1]\}.$$

Theorem 5.4 (Second-order Taylor theorem). *Let n be a positive integer, let $S \subseteq \mathbb{R}^n$ be open, and let $f : S \rightarrow \mathbb{R}$ have continuous second partial derivatives. If $\mathbf{x}, \mathbf{y} \in S$ satisfy $[\mathbf{x}, \mathbf{y}] \subseteq S$, then there is $\mathbf{w} \in [\mathbf{x}, \mathbf{y}]$ such that*

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{1}{2}(\mathbf{y} - \mathbf{x})^\top H_f(\mathbf{w})(\mathbf{y} - \mathbf{x}).$$

Proof. Define $\phi(t) := f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$. One-variable Taylor expansion from 0 to 1 gives $\phi(1) = \phi(0) + \phi'(0) + \frac{1}{2}\phi''(\theta)$ for some $\theta \in (0, 1)$. The chain rule gives

$$\phi'(0) = \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}), \quad \phi''(\theta) = (\mathbf{y} - \mathbf{x})^\top H_f(\mathbf{x} + \theta(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}).$$

Setting $\mathbf{w} := \mathbf{x} + \theta(\mathbf{y} - \mathbf{x})$ proves the result. $\square$

5.4.1 The local quadratic model

Let $S \subseteq \mathbb{R}^n$ be open, let $f : S \rightarrow \mathbb{R}$ have continuous second partial derivatives, and let $\mathbf{x} \in S$. For every sufficiently small $\mathbf{d} \in \mathbb{R}^n$, define the local quadratic model of f at $\mathbf{x}$ by

$$m_{\mathbf{x}}(\mathbf{d}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{d} + \frac{1}{2}\mathbf{d}^\top H_f(\mathbf{x})\mathbf{d}$$

and define the remainder by

$$r_{\mathbf{x}}(\mathbf{d}) := f(\mathbf{x} + \mathbf{d}) - m_{\mathbf{x}}(\mathbf{d}).$$

The second-order expansion gives

$$\frac{r_{\mathbf{x}}(\mathbf{d})}{\|\mathbf{d}\|^2} \longrightarrow 0 \quad \text{as } \mathbf{d} \rightarrow \mathbf{0}.$$

The notation

$$r_{\mathbf{x}}(\mathbf{d}) = o(\|\mathbf{d}\|^2) \quad \text{as } \mathbf{d} \rightarrow \mathbf{0}$$

is defined to mean precisely this limit. Consequently,

$$f(\mathbf{x} + \mathbf{d}) = m_{\mathbf{x}}(\mathbf{d}) + o(\|\mathbf{d}\|^2),$$

and, if $\nabla f(\mathbf{x}) = \mathbf{0}$, then

$$f(\mathbf{x} + \mathbf{d}) - f(\mathbf{x}) = \frac{1}{2}\mathbf{d}^\top H_f(\mathbf{x})\mathbf{d} + r_{\mathbf{x}}(\mathbf{d}).$$

This expansion is the bridge from curvature to the local optimality tests in the next lectures.

5.5 The Hessian test for convexity

For $\mathbf{a} \in \mathbb{R}^n$ and $\rho > 0$, define the open ball with center $\mathbf{a}$ and radius ρ by

$$B(\mathbf{a}, \rho) := \{\mathbf{z} \in \mathbb{R}^n : \|\mathbf{z} - \mathbf{a}\| < \rho\}.$$

Theorem 5.5 (Second-order characterization of convexity). *Let $C \subseteq \mathbb{R}^n$ be open and convex, and suppose $f : C \rightarrow \mathbb{R}$ has continuous second partial derivatives.*

- (a) *f is convex on C if and only if $H_f(\mathbf{x}) \succeq 0$ for every $\mathbf{x} \in C$.*
- (b) *If $H_f(\mathbf{x}) \succ 0$ for every $\mathbf{x} \in C$, then f is strictly convex on C .*
- (c) *f is concave on C if and only if $H_f(\mathbf{x}) \preceq 0$ for every $\mathbf{x} \in C$.*
- (d) *If $H_f(\mathbf{x}) \prec 0$ for every $\mathbf{x} \in C$, then f is strictly concave on C .*
- (e) *If $H_f(\hat{\mathbf{x}})$ is indefinite at one point of C , then f is neither convex nor concave on C .*

Proof. If every Hessian is PSD, Taylor's theorem gives

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}),$$

so the tangent-plane characterization proves convexity.

Conversely, suppose f is convex but there exists $\mathbf{x}^* \in C$ such that $H_f(\mathbf{x}^*)$ is not PSD. It has a negative eigenvalue λ with a unit eigenvector $\mathbf{q}$. By openness of C and continuity of the Hessian, there is an $r > 0$ such that $B(\mathbf{x}^*, 2r) \subseteq C$ and

$$\mathbf{q}^\top H_f(\mathbf{z}) \mathbf{q} < \frac{\lambda}{2} < 0 \quad \text{whenever } \mathbf{z} \in B(\mathbf{x}^*, 2r).$$

For $\mathbf{y} := \mathbf{x}^* + r\mathbf{q}$, Taylor's theorem therefore gives

$$f(\mathbf{y}) < f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^\top (\mathbf{y} - \mathbf{x}^*),$$

contradicting convexity. This proves (a). Apply (a) to $-f$ for (c). For (b), restrict f to a segment between distinct points; the resulting one-variable function has positive second derivative and is strictly convex. Parts (d) and (e) follow by the same sign arguments. $\square$

Warning

Parts (b) and (d) of [Theorem 5.5](#) are sufficient but not necessary. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) := x^4$. For distinct $x, y \in \mathbb{R}$,

$$f(y) - f(x) - f'(x)(y - x) = (y - x)^2((x + y)^2 + 2x^2) > 0.$$

The strict tangent-line inequality therefore proves that f is strictly convex, although $f''(0) = 0$. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) := -f(x)$. Then g is strictly concave although $g''(0) = 0$, so negative definiteness everywhere is likewise not necessary for strict concavity.

An indefinite Hessian somewhere proves that a function is neither convex nor concave, but the converse fails. Define $h : \mathbb{R} \rightarrow \mathbb{R}$ by $h(x) := x^3$. Since $h''(x) = 6x$ changes sign, h is neither convex nor concave on $\mathbb{R}$, even though a one-dimensional Hessian is never indefinite.

Proposition 5.6 (Quadratic functions). *Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric, let $\mathbf{b} \in \mathbb{R}^n$, and let $c \in \mathbb{R}$. Define $q : \mathbb{R}^n \rightarrow \mathbb{R}$ by*

$$q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c.$$

Then

$$q \text{ is convex} \iff \mathbf{A} \succeq 0,$$

and

$$q \text{ is strictly convex} \iff \mathbf{A} \succ 0.$$

Proof. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ be distinct, let $\lambda \in (0, 1)$, and define $\mathbf{d} := \mathbf{x} - \mathbf{y}$. Direct expansion gives

$$\lambda q(\mathbf{x}) + (1 - \lambda)q(\mathbf{y}) - q(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) = \frac{\lambda(1 - \lambda)}{2} \mathbf{d}^\top \mathbf{A} \mathbf{d}.$$

Since $\lambda(1 - \lambda) > 0$ and every nonzero vector in $\mathbb{R}^n$ can occur as $\mathbf{x} - \mathbf{y}$, nonnegativity of every Jensen gap is equivalent to $\mathbf{A} \succeq 0$, while positivity of every Jensen gap for $\mathbf{x} \neq \mathbf{y}$ is equivalent to $\mathbf{A} \succ 0$. $\square$

Warning

The quantifier “for every $\mathbf{x} \in C$ ” is essential. Semidefiniteness of the Hessian at one point does not establish convexity on C . For example, define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) := x^4 - x^2$. Then $g''(1) > 0$ but $g''(0) < 0$, so g is not convex on $\mathbb{R}$. Conversely, when the Hessian does not exist, the test is unavailable rather than negative: the convex function $r : \mathbb{R} \rightarrow \mathbb{R}$, $r(x) := |x|$, is not twice differentiable at 0.

Example 5.5. For

$$f(x_1, x_2) := 4x_1^2 + 4x_1x_2 + x_2^2, \quad H_f = \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix},$$

the eigenvalues of the Hessian are 0, 10. Thus f is convex. Since $f = (2x_1 + x_2)^2$, it is constant along a line and is not strictly convex.

Example 5.6. For

$$f(x_1, x_2, x_3) := x_1^2 + x_2^2 + x_3^2 - x_1x_2 + x_2x_3 - x_1x_3,$$

$$H_f = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}.$$

Its leading principal minors are 2, 3, 4, so $H_f \succ 0$ and f is strictly convex on $\mathbb{R}^3$.

Example 5.7 (A convex sublevel set). Let

$$S := \{\mathbf{x} \in \mathbb{R}^2 : x_1 > 1/3, \ x_1^3 + 2x_1x_2 + x_2^2 \leq 10\}.$$

On $C := \{\mathbf{x} \in \mathbb{R}^2 : x_1 > 1/3\}$, set $g(\mathbf{x}) := x_1^3 + 2x_1x_2 + x_2^2$. Its Hessian has leading minors $6x_1 > 0$ and $12x_1 - 4 > 0$, so g is strictly convex on C . Consequently, by [Proposition 3.4](#),

the sublevel set $S = \{\mathbf{x} \in C : g(\mathbf{x}) \leq 10\}$ is convex.

Chapter summary

- The Hessian records second-order directional curvature through $\mathbf{d}^\top H_f(\mathbf{x}) \mathbf{d}$.
- Eigenvalues completely characterize definiteness of symmetric matrices; leading principal minors test PD and ND, whereas all principal minors characterize PSD.
- On an open convex set, f is convex exactly when every Hessian is PSD. Pointwise PD implies strict convexity but is not necessary.
- Taylor's theorem connects local Hessian information to comparisons of function values at two points and to the local quadratic model.
- Depending on the available smoothness, convexity can be checked from the definition, the tangent-plane characterization, or the Hessian test. The Hessian condition remains necessary and sufficient for convexity; necessity is lost only in the positive-definite and negative-definite sufficient conditions for strict convexity and strict concavity.
- Convexity of a set can be proved directly, by intersection rules, or by representing the set as a sublevel set of a convex function.

5.6 Exercises

Exercise 5.1. Classify

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 5 \end{bmatrix}, \quad \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & -5 \end{bmatrix}, \quad \begin{bmatrix} -3 & 0 & 1 \\ 0 & -4 & 2 \\ 1 & 2 & -5 \end{bmatrix}.$$

Exercise 5.2. For symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$, let $f(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x}$. Compute $\nabla f(\mathbf{x})$ and $H_f(\mathbf{x})$.

Exercise 5.3. Determine the convexity of $f(\mathbf{x}) := x_1^2 + x_2^4 - 2x_1x_2$ on $D := \{\mathbf{x} \in \mathbb{R}^2 : x_2 > 1\}$.

Exercise 5.4. Determine whether $f(\mathbf{x}) := 2x_1^3 - 6x_2^2$ is convex, concave, or neither on $\mathbb{R}^2$.

Exercise 5.5. On $D := \{\mathbf{x} \in \mathbb{R}^2 : x_1 > 0, x_2 > 0\}$, show that

$$f(\mathbf{x}) := x_1^2 - 4x_1x_2 + 5x_2^2 - \log(x_1x_2)$$

is strictly convex.

Exercise 5.6. Show that $f(x, y) := x \exp(-y/x)$ is convex on $\{(x, y) \in \mathbb{R}^2 : x > 0\}$.

Exercise 5.7. Determine whether $f(\mathbf{x}) := \sqrt{x_1^2 + 2x_2^2}$ is convex, concave, or neither on $\mathbb{R}^2$.

Exercise 5.8. (a) Prove that $C := \{\mathbf{x} \in \mathbb{R}^2 : 4x_1^2 + 3x_2 \leq 9, x_2 \geq 0\}$ is convex.

(b) Determine whether $S := \{\mathbf{x} \in \mathbb{R}^3 : x_1 - x_2^2 \leq x_3 \leq x_1 + x_2^2\}$ is convex.

Exercise 5.9. Show that $D := \{\mathbf{x} \in \mathbb{R}^3 : x_1^2 + x_2^2 + 2x_3 - x_1x_2 \leq 1\}$ is convex.

Exercise 5.10. Let $S \subseteq \mathbb{R}^n$ be convex and let $h : S \rightarrow \mathbb{R}_{++}$. Deduce that $1/h$ is convex whenever h is concave.